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POWERED RATCHET TOOL WITH WIRE GUIDE INSERT

Abstract

A powered ratchet tool including a primary housing, a secondary housing coupled to the primary housing, and a motor supported within the secondary housing. The motor has an output shaft rotatable about a first axis. The powered ratchet tool further includes a ratchet mechanism supported by the secondary housing. The ratchet mechanism is operably coupled to the output shaft of the motor to be driven by the motor. Moreover, the powered ratchet tool includes an insert disposed within the secondary housing and configured to guide a plurality of wires through the secondary housing, and an output drive rotatably supported by the secondary housing. The output drive is operably coupled to the ratchet mechanism. The output drive is configured to be rotated about a second axis perpendicular to the first axis and to receive a tool element.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/745,849, filed Jan. 16, 2025, and U.S. Provisional Patent Application No. 63/557,204, filed Feb. 23, 2024, the entire contents of both of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to power tools, and more particularly to powered ratchet tools.

BACKGROUND

[0003] Powered ratchet tools may be driven in a forward direction or an opposite direction to apply torque to a fastener for tightening and loosening operations.

SUMMARY

[0004] The present disclosure provides, in one aspect, a powered ratchet tool including a primary housing, a secondary housing coupled to the primary housing, and a motor supported within the secondary housing. The motor includes an output shaft rotatable about a first axis. The powered ratchet tool further includes a ratchet mechanism supported by the secondary housing. The ratchet mechanism is operably coupled to the output shaft of the motor to be driven by the motor.

Moreover, the powered ratchet tool includes an insert disposed within the secondary housing and configured to guide a plurality of wires through the secondary housing, and an output drive rotatably supported by the secondary housing. The output drive is operably coupled to the ratchet mechanism. The output drive is configured to be rotated about a second axis perpendicular to the first axis and to receive a tool element.

[0005] The present disclosure provides, in another aspect, a powered ratchet tool including a housing having a handle housing, a gear case at least partially received within the handle housing, and a yoke housing at least partially received within the gear case. The powered ratchet tool further includes a motor supported within the gear case and a ratchet mechanism supported within the yoke housing. The handle housing, the gear case, and the yoke housing are each made of different materials.

[0006] The present disclosure provides, in another aspect, a powered ratchet tool including a housing having a handle housing, a gear case at least partially received within the handle housing, and a yoke housing at least partially received within the gear case. The powered ratchet tool further includes a motor supported within the gear case and a ratchet mechanism supported within the yoke housing. The yoke housing is partially insert-molded into the gear case.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1 is a side view of a powered ratchet tool according to an embodiment of the present disclosure.

[0008] FIG. 2 is a cross-sectional view of the powered ratchet tool of FIG. 1.

[0009] FIG. 3 is another cross-sectional view of the powered ratchet tool of FIG. 1.

[0010] FIG. 4 is a side view of the powered ratchet tool of FIG. 1 with portions removed.

[0011] FIG. 5 is a perspective view of the powered ratchet tool of FIG. 1 with portions removed,

the powered ratchet tool includes a yoke housing and an insert.

[0012] FIG. **6** is a rear view of the powered ratchet tool of FIG. **1** with portions removed.

[0013] FIG. **7** is a perspective view of the yoke housing of FIG. **5**.

[0014] FIG. **8** is a perspective view of the insert of FIG. **5**.

[0015] FIG. **9** is a perspective view of a powered ratchet tool according to another embodiment of the disclosure.

[0016] FIG. **10** is a cross-sectional view of a portion of the powered ratchet tool of FIG. **9**, taken along line **10-10** in FIG. **9**.

[0017] FIG. **11** is a cross-sectional view of the powered ratchet tool of FIG. **9**.

[0018] FIG. **12** is a perspective view of an insert of the powered ratchet tool of FIG. **9**.

[0019] FIG. **13** is a perspective view of a yoke housing of the powered ratchet tool of FIG. **9**.

[0020] FIG. **14** is a cross-sectional view of a portion of a powered ratchet tool according to another embodiment.

[0021] Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

DETAILED DESCRIPTION

[0022] With reference to FIGS. **1-3**, a powered ratchet tool **10** in accordance with an embodiment of the disclosure includes a primary housing **14** and a secondary housing or yoke housing **18** coupled to the primary housing **14** by fasteners **101** (FIG. **6**). A portion of the yoke housing **18** is received within the primary housing **14**. The primary housing **14** is a clamshell housing configured to be grasped by a user during operation, while the yoke housing **18** is a metal housing. The yoke housing **18** has a first end **20a** disposed within the primary housing **14** and a second end **20b** extending from the primary housing **14** (FIG. **7**). The powered ratchet tool **10** further includes a motor **22** that is supported within the yoke housing **18**. The motor **22** has an output shaft **26** rotatable about a first axis **30** and is configured to provide torque to an output drive **34** rotatably supported by the yoke housing **18** for rotation about a second axis **36** perpendicular to the first axis **30**. The motor **22** is preferably a brushless DC motor. In some embodiments, the motor **22** is a surface permanent magnet (SPM) motor including a stator, a rotor, and permanent magnets affixed to or embedded in an exterior surface of the rotor. In other embodiments, the motor **22** is an outer rotor motor, having a rotor that surrounds and rotates about the stator. The output drive **34** includes a square drive interface **40**, preferably with a nominal size (e.g., $\frac{3}{8}$ ", $\frac{1}{2}$ ", $\frac{3}{4}$ ", 1 ", etc.). The output drive **34** is configured to be coupled to a tool element (not shown) that performs work on a workpiece (e.g., a fastener, bit, or the like).

[0023] In the illustrated embodiment, the ratchet tool **10** includes a battery pack **38** received by a battery receptacle **42** formed in the primary housing **14** opposite the yoke housing **18**. The battery receptacle **42** electrically connects the battery pack **38** to the motor **22** (via suitable electrical and electronic components, such as a PCBA containing MOSFETs, IGBTs, or the like). The battery pack **38** may be a 12-volt power tool battery pack that includes three lithium-ion battery cells. Alternatively, the battery pack **38** may include fewer or more battery cells to yield any of a number of different output voltages (e.g., 14.4 volts, 18 volts, etc.). Additionally, or alternatively, the battery cells may include chemistries other than lithium-ion such as, for example, nickel cadmium, nickel metal-hydride, or the like. The ratchet tool **10** also includes an actuator **44** for controlling operation of the ratchet tool **10** (e.g., to energize/de-energize the motor **22**). In the illustrated embodiment, the actuator **44** is a push-button that can be depressed into the primary housing **14** to energize the motor **22**. The illustrated actuator **44** extends from the primary housing **14** in the same direction as the output drive **34**.

[0024] With reference to FIGS. 2 and 3, the powered ratchet tool **10** further includes a ratchet mechanism **45** supported by the second end **20b** of the yoke housing **18**. The ratchet mechanism **45** has a crankshaft **46** with an eccentric member **48** and a drive bushing **50** arranged on the eccentric member **48** of the crankshaft **46**. The crankshaft **46** is rotatably supported by two bearings **47** fixed within the yoke housing **18**. The ratchet mechanism **45** further includes a yoke **54** through which the output drive **34** extends. The yoke **54** has a recess **58** in which the drive bushing **50** is arranged. When the crankshaft **46** is rotated, the drive bushing **50** pivots the yoke **54** in a reciprocating manner to drive the output drive **34**.

[0025] With continued reference to FIGS. 2 and 3, the crankshaft **46** of the ratchet mechanism **45** is operably coupled to the output shaft **26** of the motor **22** via a transmission **62** also supported within the yoke housing **18** to be driven by the motor **22**. The transmission **62** includes a first planet carrier **66** extending through a portion of the motor **22** and surrounding the output shaft **26** of the motor **22**, a pinion **68**, a ring gear **70** coupled to the first planet carrier **66**, a plurality of planet gears **74**, and a second planet carrier **78**. The pinion **68** is coupled to the output shaft **26** of the motor **22** for rotation about the first axis **30**. The plurality of planet gears **74** are meshed with both the pinion **68** and the ring gear **70**. Pins **82** are used to couple the plurality of planet gears **74** with the second planet carrier **78**. The second planet carrier **78** engages the crankshaft **46** to drive rotation of the crankshaft **46** to operate the ratchet mechanism **45**.

[0026] With reference to FIGS. 4-8, the powered ratchet tool **10** also includes an insert **86** disposed at the first end **20a** of the yoke housing **18**. The insert **86** is formed of a plastic material in the illustrated embodiment but may be made of other materials. The insert **86** has a first end **88a** and a second end **88b** opposite the first end **88a**. The insert **86** defines a first channel **90a** and a second channel **90b** that are opposite each other and extend between the first and second ends **88a**, **88b**. The first and second channels **90a**, **90b** of the insert **86** are configured to provide support at the first end **20a** of the yoke housing **18** for a first plurality of wires **102a** and a second plurality of wires **102b** extending between two PCBs **106a**, **106b** and the motor **22**. The first and second channels **90a**, **90b** extend in a direction parallel to the first axis **30**, and thereby provides a path for the first and second plurality of wires **102a**, **102b** to guide the wires **102a**, **102b** through the yoke housing **18**.

[0027] With reference to FIG. 8, the insert **86** also includes a plurality of channel arms **92** and an end portion **94**. The channel arms **92** are coupled to and extend between the first channel **90a** and the second channel **90b** to interconnect the channels **90a**, **90b** together. Each channel arm **92** has an arcuate shape so that the insert **86** is shaped and sized to fit within the yoke housing **18**. The channel arms **92** are spaced from each other such that a gap is formed between adjacent channel arms **92**. The end portion **94** is coupled to the first channel **90a** and the second channel **90b** at the first end **88a** of the insert **86**. As such, the insert **86** defines a space **96** between the first channel **90a**, the second channel **90b**, the plurality of channel arms **92**, and the end portion **94**. The motor **22** is configured to be received within the space **96** to be supported by the insert **86**, and in turn, supported by the yoke housing **18**. An impeller **97**, that is provided for cooling the motor **22**, is arranged rearward of the motor **22** and received within the space **96** of the insert **86**.

[0028] In addition, the insert **86** includes multiple fastening holes **98** defined within the end portion **94**. It should be assumed that the insert **86** has multiple fastening holes **98**, even though only one fastening hole **98** is illustrated in FIG. 8. More specifically, each fastening hole **98** is defined within a respective fastener arm **100** formed on the end portion **94** of the insert **86**. The fastener arms **100** are configured to receive the fasteners **101** that couple the yoke housing **18** coupled to the primary housing **14**.

[0029] The yoke housing **18** defines a first window **110a** along a top portion **114** of the yoke housing **18** and a second window **110b** along a bottom portion **118** of the yoke housing **18**. The insert **86** is arranged within the yoke housing **18** such that the first channel **90a** is exposed through the first window **110a** of the yoke housing **18** and the second channel **90b** is exposed through the

second window **110b** of the yoke housing **18**. The first and second channels **90a**, **90b** of the insert **86** are shaped and sized to respectively fit within the first and second windows **110a**, **110b** of the yoke housing **18**.

[0030] In operation, the user depresses the actuator **44** to energize the motor **22** and rotate the output shaft **26**. The pinion **68** co-rotates with the output shaft **26** to rotate the plurality of planet gears **74**, and thereby rotate the second planet carrier **78** to drive rotation of the crankshaft **46**. The crankshaft **46** rotates the drive bushing **50**, which causes the yoke **54** to pivot in a reciprocating manner relative to the yoke housing **18**. The yoke **54** then transfers torque to the output drive **34** to either tighten or loosen a workpiece.

[0031] FIGS. **9-11** illustrated a powered ratchet tool **210** according to another embodiment. The powered ratchet tool **210** is similar in some aspects to the powered ratchet tool **10** of FIGS. **1-8** described above, and features of the powered ratchet tool **210** corresponding with features of the powered ratchet tool **10** are given like reference numerals plus '200.' In addition, the following description focuses primarily on differences between the powered ratchet tool **210** and the powered ratchet tool **10**, and it should be understood that features of the powered ratchet tool **10** may be incorporated into the powered ratchet tool **210**, and vice versa.

[0032] With reference to FIGS. **9-12**, the powered ratchet tool **210** includes a housing **212**, which, in the illustrated embodiment, includes a handle housing **214**, a gear case **216**, and a yoke housing **218**. A portion of the gear case **216** is received within the handle housing **214**, and a portion of the yoke housing **218** is received within the gear case **216**.

[0033] Referring to FIG. **10**, a motor **222** is supported within the gear case **216**. The motor **222** has an output shaft **226** rotatable about a first axis **230** and is configured to provide torque to an output drive **234** rotatably supported by the yoke housing **218** for rotation about a second axis **236** perpendicular to the first axis **230**. The motor **222** is an outer rotor motor in the illustrated embodiment, having a rotor that surrounds and rotates about the stator. The output drive **234** includes a square drive interface **240** but may have other drive interfaces in other embodiments. The output drive **234** is configured to be coupled to a tool element (e.g., a socket; not shown) that performs work on a workpiece (e.g., a fastener, bit, or the like). In further embodiments, the output drive **234** may include an internal recess having a polygonal cross-sectional shape and configured to receive the tool element. In additional embodiments, the output drive **234** and the tool element may include cooperating spline geometries. The ratchet tool **10** also includes a battery pack **238** received by a battery receptacle **242** formed in the handle housing **214** opposite the yoke housing **218**. An actuator **244** is provided for controlling operation of the ratchet tool **210** (e.g., to energize/de-energize the motor **222**) and is a push-button that can be depressed into the handle housing **214** to energize the motor **222**.

[0034] With continued reference to FIG. **10**, the powered ratchet tool **210** includes a ratchet mechanism **245** supported by the yoke housing **218**. The ratchet mechanism **245** has a crankshaft **246** with an eccentric pin **248** and a drive bushing **250** arranged on the eccentric pin **248**. The crankshaft **246** is rotatably supported by two bearings **247** fixed within the yoke housing **218**. The ratchet mechanism **245** further includes a yoke **254** through which the output drive **234** extends. The yoke **254** has a recess **258** in which the drive bushing **250** is arranged. When the crankshaft **246** is rotated, the drive bushing **250** pivots the yoke **254** in a reciprocating manner to drive the output drive **234**.

[0035] With continued reference to FIG. **10**, the crankshaft **246** of the ratchet mechanism **245** is operably coupled to the output shaft **226** of the motor **222** via a transmission **262** supported within the gear case **216** to be driven by the motor **222**. The transmission **262** includes a planet carrier **266**, a pinion **268**, a ring gear **270**, and a plurality of planet gears **274**. The pinion **268** is coupled to the output shaft **226** of the motor **222** for rotation about the first axis **230**. The plurality of planet gears **274** are meshed with both the pinion **268** and the ring gear **270**. The ring gear **270** is fixed within the gear case **216** (e.g., via a press-fit or any other suitable method, such as insert-molding).

The planet carrier **266** engages the crankshaft **246** (e.g., via a spline connection, press-fit, or any other suitable connection) to drive rotation of the crankshaft **246** to operate the ratchet mechanism **245**.

[0036] With reference to FIGS. **11** and **12**, The powered ratchet tool **210** also includes an insert **286** formed of a plastic material in the illustrated embodiment but may be made of other materials. The insert **286** has a first end **288a** and a second end **288b** opposite the first end **288a**. The insert **286** defines a channel **290** arranged opposite a flat surface **322** formed on the gear case **216**. The channel **290** extends between the first and second ends **288a**, **288b** of the insert **286**. The channel **290** of the insert **286** and the flat surface **322** of the gear case **216** are configured to provide support for a first plurality of wires **302a** and a second plurality of wires **302b** extending between two PCBs **306a**, **306b** and the motor **222**. The channel **290** and the flat surface **322** extend in a direction parallel to the first axis **230**, and thereby provide a path for the first and second plurality of wires **302a**, **302b** to guide the wires **302a**, **302b** through and along the gear case **216**. In other embodiments, all or a portion of the insert **286** may be formed of a material (e.g., aluminum) that provides additional strength in comparison to the plastic material. As such, at least a portion of the insert **286** in such embodiments is stiffer than the material of the handle housing **214**. This portion of the insert **286** may be separate from or integral with the channel **290**.

[0037] The housing **212** of the powered ratchet tool **210** is constructed for improved strength and rigidity compared to typical powered ratchet tools, which often include a metal yoke housing coupled to a plastic handle housing. This in turn may enhance the ability of the powered ratchet tool **210** to manually apply torque to a fastener by grasping and rotating the housing **212** about the second axis **236**.

[0038] The illustrated handle housing **214** is a clamshell housing configured to be grasped by a user during operation and is made of plastic. The gear case **216** and the yoke housing **218** are made of metal. For example, in the illustrated embodiment, the gear case **216** is made of aluminum (e.g., by die casting). The yoke housing **218** is made of steel (e.g., by machining, forging, or die casting). Thus, in the illustrated embodiment, handle housing **214**, the gear case **216**, and the yoke housing **218** are each made of different materials.

[0039] The yoke housing **218** is insert-molded into the gear case **216** in the illustrated embodiment. That is, the gear case **216** is die cast around the yoke housing **218** to couple the yoke housing **218** to the gear case **216** with a strong and fastener-free connection. In the illustrated embodiment, the yoke housing **218** includes a tubular neck **326** with a knurled texture **330** on its outer surface (FIG. **13**). The knurled texture **330** may provide increased surface area and bonding between the gear case **216** and the yoke housing **218**.

[0040] FIG. **14** illustrates another embodiment of a yoke housing **418** and a gear case **416** which may be incorporated into the powered ratchet tool **210**. The illustrated yoke housing **418** includes a bushing **534** having a knurled texture on its outer surface. The bushing **534** is separate from the remainder of the yoke housing **418** and is insert-molded into the gear case **416**. The bushing **534** may be made of steel. Once the bushing **534** is molded into the gear case **416**, the remainder of the yoke housing **418** (e.g., a body or the tubular neck **526** of the yoke housing **418**) is pressed into the bushing **534**. As such, the tubular neck **526** of the illustrated yoke housing **418** has a smooth outer surface. By providing the knurled texture on the bushing **534** rather than the body of the yoke housing **418**, manufacturing of the yoke housing **418** may be simplified. In addition, because only the bushing **534** is insert-molded into the gear case **416**, the mold may be simplified.

[0041] Although the disclosure has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the disclosure as described.

[0042] Various features and aspects of the disclosure are set forth in the following claims.

Claims

1. A powered ratchet tool comprising: a primary housing; a secondary housing coupled to the primary housing; a motor supported within the secondary housing, the motor including an output shaft rotatable about a first axis; a ratchet mechanism supported by the secondary housing, the ratchet mechanism operably coupled to the output shaft of the motor to be driven by the motor; an insert disposed within the secondary housing and configured to guide a plurality of wires through the secondary housing; and an output drive rotatably supported by the secondary housing and operably coupled to the ratchet mechanism, the output drive configured to be rotated about a second axis perpendicular to the first axis and to receive a tool element.
2. The powered ratchet tool of claim 1, wherein the insert is formed of a plastic material.
3. The powered ratchet tool of claim 1, wherein the insert defines a channel extending in a direction parallel to the first axis and configured to guide the plurality of wires through the secondary housing.
4. The powered ratchet tool of claim 3, wherein the secondary housing defines a window, and wherein the channel is exposed through the window.
5. The powered ratchet tool of claim 3, wherein the channel is a first channel and the plurality of wires is a first plurality of wires, and wherein the insert defines a second channel opposite the first channel and configured to guide a second plurality of wires through the secondary housing.
6. The powered ratchet tool of claim 5, wherein the secondary housing defines a first window and a second window, and wherein the first channel is exposed through the first window and the second channel is exposed through the second window.
7. The powered ratchet tool of claim 1, wherein the insert is formed of aluminum.
8. The powered ratchet tool of claim 1, further comprising a transmission supported by the secondary housing and configured to operably couple the ratchet mechanism to the output shaft of the motor.
9. A powered ratchet tool comprising: a housing including a handle housing, a gear case at least partially received within the handle housing, and a yoke housing at least partially received within the gear case; a motor supported within the gear case; and a ratchet mechanism supported within the yoke housing, wherein the handle housing, the gear case, and the yoke housing are each made of different materials.
10. The powered ratchet tool of claim 9, wherein the handle housing is made of plastic, and wherein the gear case and the yoke housing are made of metal.
11. The powered ratchet tool of claim 10, wherein the gear case is made of aluminum and the yoke housing is made of steel.
12. The powered ratchet tool of claim 9, further comprising an insert disposed within the gear case and configured to guide a plurality of wires through the gear case.
13. The powered ratchet tool of claim 12, wherein the insert defines a channel configured to guide the plurality of wires.
14. The powered ratchet tool of claim 9, wherein the yoke housing has a tubular neck with a knurled texture.
15. The powered ratchet tool of claim 9, further comprising a transmission supported by the gear case and configured to operably couple the ratchet mechanism to the motor.
16. A powered ratchet tool comprising: a housing including a handle housing, a gear case at least partially received within the handle housing, and a yoke housing at least partially received within the gear case; a motor supported within the gear case; and a ratchet mechanism supported within the yoke housing, wherein the yoke housing is partially insert-molded into the gear case.
17. The powered ratchet tool of claim 16, wherein the yoke housing includes a bushing insert-molded into the gear case and a body pressed into the bushing.

18. The powered ratchet tool of claim 17, wherein the bushing has an outer surface with a knurled texture.

19. The powered ratchet tool of claim 17, wherein the bushing is made of steel.

20. The powered ratchet tool of claim 16, further comprising an insert disposed within the gear case and configured to guide a plurality of wires through the gear case.
